

Name: Srinidhi A  
ID: COMETFWC053

## 7-Segment Display Analysis

### Question 59

Two products are sold from a vending machine with push buttons  $P_1$  and  $P_2$ .

- No button  $\rightarrow$  display 0
- Only  $P_1 \rightarrow$  display 2
- Only  $P_2 \rightarrow$  display 5
- Both pressed  $\rightarrow$  display E

Pressed = 1, Not pressed = 0. Segment ON = 1.

### Truth Table

| $P_1$ | $P_2$ | Output |
|-------|-------|--------|
| 0     | 0     | 0      |
| 1     | 0     | 2      |
| 0     | 1     | 5      |
| 1     | 1     | E      |

### 7-Segment Pattern

| Char | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ | $g$ |
|------|-----|-----|-----|-----|-----|-----|-----|
| 0    | 1   | 1   | 1   | 1   | 1   | 1   | 0   |
| 2    | 1   | 1   | 0   | 1   | 1   | 0   | 1   |
| 5    | 1   | 0   | 1   | 1   | 0   | 1   | 1   |
| E    | 1   | 0   | 0   | 1   | 1   | 1   | 1   |

## Logic Expressions

$$a = 1$$

$$b = \overline{P_2}$$

$$c = \overline{P_1}$$

$$d = 1$$

$$e = P_1 + \overline{P_2}$$

$$f = \overline{P_1} + P_2$$

$$g = P_1 + P_2$$

## Conclusion

The required segment outputs are derived from the button combinations. These expressions can be implemented using logic gates or programmed into a controller for hardware realization.